Instructor Notes

Pedro Pathing 3 for FIRST Tech Challenge · Version 2
Companion to **FTC_PedroPathing_v2.pdf** (16 slides)

> **How to use these notes.** One entry per slide, in order. Each entry gives the slide title, a suggested time, what to say, a question to ask the class, and anything to watch out for. Times are a guide for a first delivery — total them up and plan your sessions around the natural break points marked in the notes.

> **You do not need to be a programmer to deliver this.** Everything on the slides is sourced and cited, and the notes tell you what the point of each slide is. Where a question has a right answer, the answer is given in brackets. If a student asks something the notes do not cover, “I don't know, let's look it up” is a good answer — and modelling that is part of the lesson.

> **The single best thing you can do** is run the questions. Students who have said an idea out loud remember it; students who have only heard it do not. Where a note says to demonstrate something physically or on a whiteboard, that is worth more than the extra minutes it costs.

Opening

1 Title slide

Opening. 1 min.

Say: Check the room has done the 101 deck. This one assumes objects, methods and the loop are already familiar. If they are not, stop and do 101 first.

Ask the class: Ask: who can tell me what loop() does? If nobody can, do not start this deck today.

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2 Two stops. Then you can read our drive code.

2 min. The warning is the important half.

Say: Say the Pedro 2 warning loudly and early. Students will search YouTube the moment they get stuck, and almost everything they find will be for the old version. Give them the habit of checking the version on every page they open.

Ask the class: Ask: you find a tutorial that uses followPath(). How do you check if it is current? (Look for Pedro 3 in the title or on the page; compare against the official docs.)

Part 1 — Knowing where you are

3 Section opener — Knowing where you are

Section opener. 1 min.

Say: Short section. Three real ideas.

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4 Eight words you are about to meet.

Preview. 2 min.

Say: Read and move on.

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5 Pedro answers two questions, over and over.

4 min.

Say: The taxi analogy does the work: satnav plus driver. Students often think Pedro is only for autonomous — say clearly that it drives in TeleOp too, which Part 2 shows.

Ask the class: Ask: which half is broken if the robot drives smoothly but ends up in the wrong place? (The localizer — it thinks it is somewhere it is not.)

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6 A Pose is where you stand plus which way you face.

5 min.

Say: Stand up and demonstrate: walk to a spot in the room (x and y), then turn on the spot (heading). Same position, different pose. That physical demo lands better than any diagram. The units warning is a real trap — say it twice.

Ask the class: Ask: I stand in the corner facing the door, then turn to face the window. Did my x and y change? (No.) Did my pose change? (Yes.)

Watch out: Do not go into why Pedro uses its own coordinate system. Just tell them it does and point at the Coordinates page.

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7 Build it in init(). Feed it in loop(). Every loop.

6 min. The most important slide in this deck.

Say: Everything in Programming 101 Part 3 pays off here. A blocking loop does not just freeze the drive — it makes Pedro lose the robot's position, and the robot will then drive confidently to the wrong place. Say that consequence out loud.

Ask the class: Ask: what happens if I put a two-second wait inside loop()? (Pedro misses 100 updates and its idea of where we are drifts badly.)

Watch out: If students remember one line from this deck, it is follower.update() every loop. Write it on the whiteboard and leave it there.

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8 Eight words you now own.

Recap. 2 min.

Say: Cover and quiz.

Part 2 — Making it move

9 Section opener — Making it move

Section opener. 1 min.

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10 Ten words you are about to meet.

Preview. 2 min.

Say: Read and move on.

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11 A path is a line or a curve between two Poses.

5 min. Open the visualizer live if you have a projector and wifi.

Say: Five minutes of dragging a control pose around in the visualizer teaches curves better than any explanation. Point out that the robot's heading and its direction of travel are two separate things — that is what interpolation controls.

Ask the class: Ask: we need to drive around the submersible without hitting it. Line or curve? (Curve, with a control pose pushed out to the side.)

Watch out: Do not go deep on Bezier maths. 'The control pose pulls the line towards it' is the right level for this audience.

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12 Pedro still drives. You just hold the sticks.

5 min.

Say: Demonstrate robot-centric versus field-centric physically. Stand facing the class and push forward — you move away from them. Now turn 180 degrees and push forward again: robot-centric moves you towards them, field-centric would still move you away. Drivers strongly prefer field-centric once they try it.

Ask the class: Ask: the robot is facing the driver and you push the stick forward. Which way does it go, robot-centric? (Towards you.) Field-centric? (Away from you.)

Watch out: Note that follower.update() appears here too. It is not optional in TeleOp.

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13 Tuning is measuring your robot. Not guessing numbers.

5 min.

Say: The framing that matters: tuning is measurement, not trial and error. Each step depends on the one before, so doing them out of order wastes an afternoon. Book a dedicated session with the real robot for this — it cannot be taught from slides.

Ask the class: Ask: why can Pedro not just work out our robot's weight by itself? (It has no scale. It only sees motor commands and wheel rotation.)

Watch out: Do not attempt live tuning in a lecture. Schedule it separately with the robot, a laptop and two hours.

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14 You are not the first person to be stuck on this.

3 min.

Say: Make the evidence rule a team norm. 'It does not work' gets no useful answer anywhere, including on Discord. The four things in the footer are what turns a complaint into a question someone can answer.

Ask the class: Ask: what four things do you bring when you ask for help?

Watch out: An adult should supervise any Discord use, per your team's safeguarding rules.

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15 Ten words you now own.

Recap. 2 min.

Say: Cover and quiz.

Close

16 Cheat sheet — five sentences.

2 min.

Say: Five sentences is the honest takeaway from one session on Pedro. Tell them the real learning happens with the robot on the floor, and that this deck exists so they know what the words mean when that day comes.

Watch out: Book the tuning session before everyone leaves the room.